Cryptocurrency Data Schema: Updated

Entity-Relationship Diagrams

This document provides the updated Entity-Relationship Diagrams (ERDs) for the Cryptocurrency Data Schema, reflecting the critical architectural change to a multi-database system and the latest schema modifications.

Architectural Change: Multi-Database Split

The system is now split into two distinct databases to optimize for different data access patterns and persistence requirements. All diagrams below indicate the database location for each table.

Database Name	Type	Purpose
`metadata_db`	PostgreSQL	Stores static, low-volume metadata (e.g., token definitions).
`crypto_db`	TimescaleDB	Stores high-volume, time-series data (e.g., OHLCV, trades, metrics, and analysis).

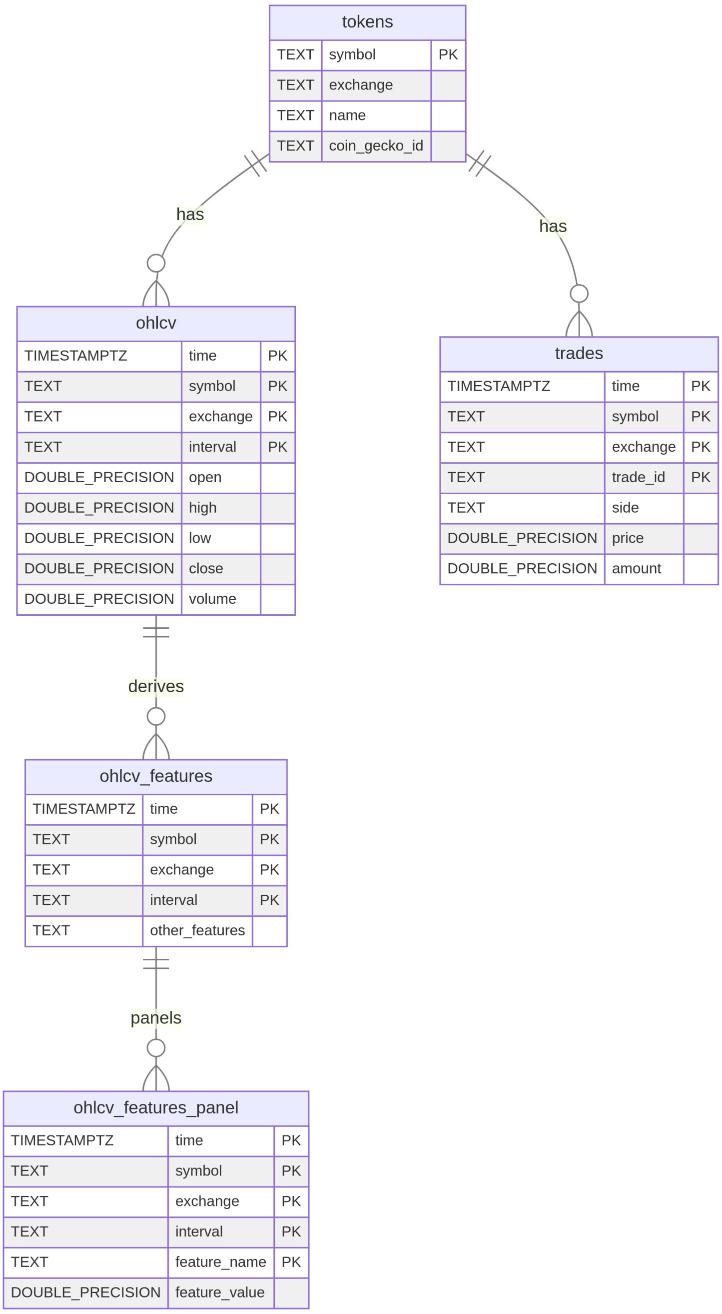
Section 1: Market & Feature Data

This section focuses on the core market data and the feature engineering pipeline.

Table Schemas

Table Name	Database	Ingestion Source / API	Matches Project Architecture
`tokens`	PostgreSQL	CoinGecko API (master token metadata)	Forecasting Module
`ohlcv`	TimescaleDB	Binance (CCXT)	Forecasting Module
`trades`	TimescaleDB	Exchange APIs (CCXT)	Forecasting Module
`ohlcv_features`	TimescaleDB	Derived internally from `ohlcv` via CoinPreprocessor	ML Panel (sarimax/prophet)
`ohlcv_features_panel`	TimescaleDB	Derived internally from `ohlcv_features` via PanelPreprocessor	ML Panel (CNNLSTM, TFT)

Entity-Relationship Diagram (Market & Feature Data)



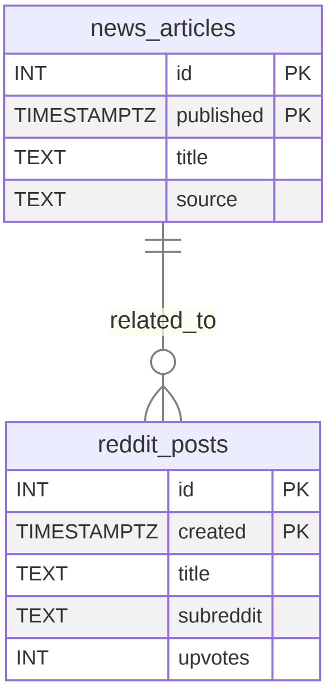
Section 2: Sentiment & News Data

This section covers alternative data sources for sentiment and news analysis.

Table Schemas

Table Name	Database	Ingestion Source / API	Matches Project Architecture
`news_articles`	TimescaleDB	CryptoPanic News API	RAG Module
`reddit_posts`	TimescaleDB	Reddit API	RAG Module

Entity-Relationship Diagram (Sentiment & News Data)



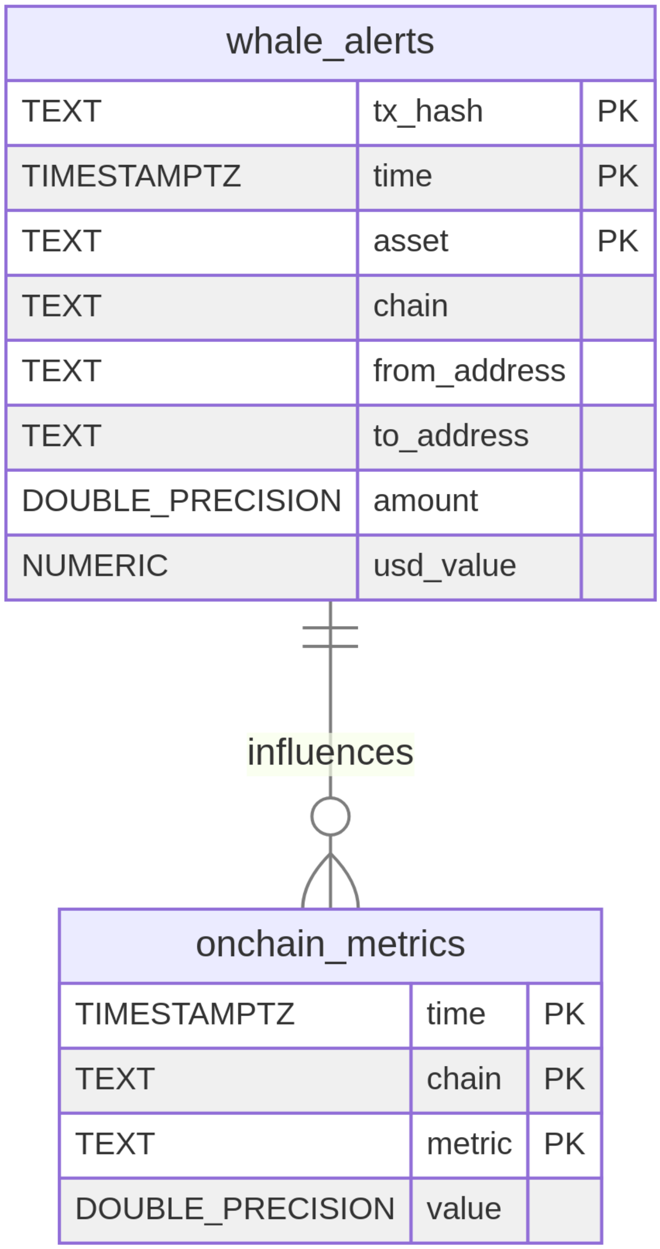
Section 3: On-Chain Analytics

This section focuses on tables related to blockchain monitoring and metrics.

Table Schemas

Table Name	Database	Ingestion Source / API	Matches Project Architecture
`whale_alerts`	TimescaleDB	Alchemy (blockchain transaction ingestion)	On-Chain Module
`onchain_metrics`	TimescaleDB	Derived from blockchain RPCs (Infura metrics aggregation)	On-Chain Module

Entity-Relationship Diagram (On-Chain Analytics)



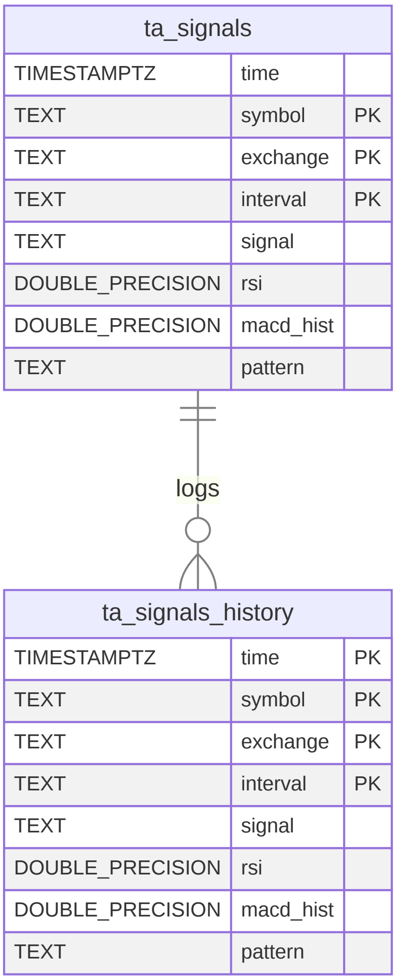
Section 4: Analysis

This section contains tables used for storing the results of technical analysis, specifically the generated signals and their history. All tables are located in the `crypto_db`.

Table Schemas

Table Name	Database	Ingestion Source / API	Matches Project Architecture
`ta_signals`	TimescaleDB	Derived internally from `ohlcv_features`	Technical Analysis Module
`ta_signals_history`	TimescaleDB	Internal (Log of `ta_signals` changes)	Technical Analysis Module

Entity-Relationship Diagram (Analysis)



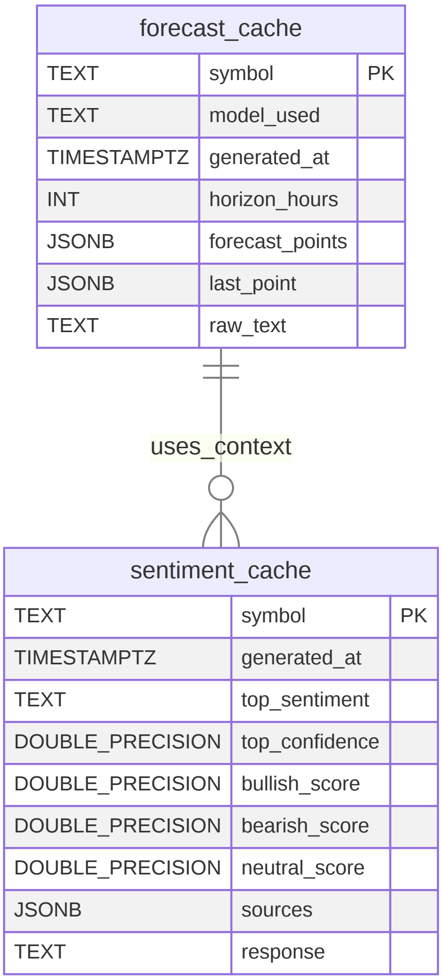
Section 5: Caching

This section contains tables used exclusively for caching the results of forecasting and sentiment processing to ensure fast dashboard loading. All tables are located in the `crypto_db`.

Table Schemas

Table Name	Database	Ingestion Source / API	Matches Project Architecture
`forecast_cache`	`TimescaleDB`	Internal (Caching of ML model results)	Forecasting Module
`sentiment_cache`	`TimescaleDB`	Internal (Caching of Sentiment Analysis results)	Sentiment Module

Entity-Relationship Diagram (Caching)



Section 6: System Management

This section covers tables used for pipeline monitoring and system state tracking.

Table Schemas

Table Name	Database	Ingestion Source / API	Matches Project Architecture
`ingestion_jobs`	TimescaleDB	Internal – updated by ingestion pipelines for every data source	Pipeline Monitoring

Entity-Relationship Diagram (System Management)

ingestion_jobs		
TEXT	pipeline	PK
TIMESTAMPTZ	start_time	
TIMESTAMPTZ	end_time	
TEXT	status	
TEXT	source	